compute the locations of $\mathcal{M}$'s head at each time step. This results in the internal representations of the form

$$\mathbf{x}_t^1 = \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} \tag{74}$$

for $t \in \mathbb{N}_{\geq 0}$, where $c(t)$ denotes the position of $\mathcal{M}$'s head at time t. This component is identical to the *second* layer of the transformer from Pérez et al. (2021) and is explained in more detail in Lemma F.6.

3. **Layer 2** ($\mathcal{L}_2$; Lemma F.7): Uses the $\mathcal{L}_1$-computed information about the head locations at each time step to (almost) compute s_t—the symbol read by the head of the PTM at time step t. This results in the internal representations of the form

$$\mathbf{x}_t^2 = \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ \ell(t), [\![v_{\ell(t)}]\!] \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} \tag{75}$$

for $t \in \mathbb{N}_{\geq 0}$, where $\ell(t)$ denotes the last time step when $\mathcal{M}$'s head wrote to $c(t)$. This component is analogous to the *third* layer of the transformer from Pérez et al. (2021) and is explained in more detail in Lemma F.7.

4. **Output function** (F; Lemma F.8): The function F uses the information computed by the two layers of the transformer to compute the one-hot encoding of the current configuration of the PTM. In particular, this includes $\mathcal{M}$'s current state (q_t), the symbol read by $\mathcal{M}$'s head (s_t), and, for reasons that we explain shortly, the action performed by $\mathcal{M}$ at the previous time step (a_{t-1}). This results in the representation

$$\mathsf{enc}\left(\boldsymbol{y}_{<t}^{\text{💬}}\right) = [\![q_t, s_t, a_{t-1}]\!], \tag{76}$$

which is used for sampling the next transition of the PTM. This component resembles the output function of Pérez et al. (2021) and is explained in more detail in Lemma F.8.

5. **Sampling** (Lemma F.9): The representation $[\![q_t, s_t, a_{t-1}]\!]$ is used to index the output matrix $\mathbf{E}$ that contains the conditional probabilities $p(\cdot \mid q_t, s_t)$. This can be used to sample the next augmented symbol $[\![q^{t+1}, v_t, \overline{y}_t, a_t, a_{t-1}]\!]$ (here, the last component, a_{t-1}, equals the "input" to the sampling step). This component is explained in more detail in Lemma F.9.

The idea of the construction—iteratively computing and storing the modifications to the PTM configuration in the generated string $\boldsymbol{y}^{\text{💬}}$—is therefore identical to Pérez et al.'s (2021) one. In particular, the two components that perform the bulk of the simulation—Layer 1 and Layer 2—are identical to the components from Pérez et al. (2021). We describe and show the correctness of the components in the lemmata in the rest of the section. The correctness of the construction follows from the correctness of the components.

Weak equivalence. To define a CoT-augmented LM with such a transformer, we can define a transducer that projects the outputs in $\left(Q \times \Gamma \times \overline{\Sigma}_\varepsilon \times A \times A\right)^*$ onto Σ^* in the standard way by retaining only the outputs in $\overline{\Sigma}_\varepsilon$. This collapses all the executions of the transformer LM (which simulates the executions of the PTM with the same probabilities, as per Lemma F.9) yielding the same string in Σ^*, resulting in a CoT-augmented transformer LM weakly equivalent to p.

Precision. The constructed transformer computes and stores position-dependent values at several points during the computation. The precision required for the representation of these values grows logarithmically with the number of computational steps. However, since the number of computational steps performed by a PTM generating a string is potentially unbounded in the length of the string (Hopcroft et al., 2001, p. 339), the transformer's precision is subsequently unbounded as well (Nowak et al., 2023). ■

We begin with a general lemma about the disjunction of one-hot encodings.

Lemma F.4. *Let $\mathcal{S}_1, \ldots, \mathcal{S}_n$ be finite sets and let $[\![s_1, \ldots, s_n]\!] \in \{0,1\}^{|\mathcal{S}_1| \cdots |\mathcal{S}_n|}$ denote the one-hot encoding of the tuple $(s_1, \ldots, s_n) \in \mathcal{S}_1 \times \cdots \times \mathcal{S}_n$. Define the matrices*

$$\mathbf{W}^{\mathcal{S}_i} \in \{0,1\}^{|\mathcal{S}_i| \times |\mathcal{S}_1| \cdots |\mathcal{S}_n|} \tag{77}$$

element-wise as

$$\mathbf{W}^{\mathcal{S}_i}_{s,(s_1,\ldots,s_{i-1},s,s_{i+1},\ldots,s_n)} \stackrel{\text{def}}{=} 1 \qquad \text{for all } s_j \in \mathcal{S}_j, j \in 1, \ldots, n, j \neq i. \tag{78}$$

Then, it holds that

$$\mathbf{W}^{\mathcal{S}_i} [\![s_1, \ldots, s_n]\!] = [\![s_i]\!]. \tag{79}$$

Proof. Eq. (78) can equivalently be written as

$$\mathbf{W}^{\mathcal{S}_i}_{s,(s_1,\ldots,s_{i-1},s_i,s_{i+1},\ldots,s_n)} \stackrel{\text{def}}{=} \mathbb{1}\{s_i = s\}. \tag{80}$$

Indexing the elements of $\mathbf{W}^{\mathcal{S}_i} [\![s_1, \ldots, s_n]\!]$ directly with $s \in \mathcal{S}_i$, we then compute

$$\begin{aligned}
\left(\mathbf{W}^{\mathcal{S}_i} [\![s_1, \ldots, s_n]\!]\right)_s &= \sum_{\substack{s'_j \in \mathcal{S}_j \\ j=1,\ldots,n}} \mathbf{W}^{\mathcal{S}_i}_{s,(s'_1,\ldots,s'_n)} [\![s_1, \ldots, s_n]\!]_{s'_1,\ldots,s'_n} && \text{(81a)}\\
&= \sum_{\substack{s'_j \in \mathcal{S}_j \\ j=1,\ldots,n}} \mathbb{1}\{s'_i = s\} [\![s_1, \ldots, s_n]\!]_{s'_1,\ldots,s'_n} && \text{(81b)}\\
&= \sum_{\substack{s'_j \in \mathcal{S}_j \\ j=1,\ldots,n}} \mathbb{1}\{s'_i = s\} \mathbb{1}\{s'_1 = s_1, \ldots, s'_n = s_n\} && \text{(81c)}\\
&= \sum_{s'_i \in \mathcal{S}_i} \mathbb{1}\{s'_i = s\} \underbrace{\sum_{\substack{s'_j \in \mathcal{S}_j \\ j=1,\ldots,n,\ j \neq i}} \mathbb{1}\{s'_1 = s_1, \ldots, s'_n = s_n\}}_{=1} && \text{(81d)}\\
&= \sum_{s'_i \in \mathcal{S}_i} \mathbb{1}\{s'_i = s\}, && \text{(81e)}
\end{aligned}$$

which is the definition of the elements of $[\![s_i]\!]$. The equality in Eq. (81d) follows from the fact that the summand is non-zero exactly when $s'_j = s_j$ for all $j \neq i$. ■

Lemma F.5 (Input representations)**.** *Define the following static representation function of the CoT-augmented symbols $y^{💬} \stackrel{\text{def}}{=} (q, v, \underline{\overline{y}}, a', a) \in \Delta$:*

$$\mathbf{r}_\Pi\left(y^{💬}, t\right) \stackrel{\text{def}}{=} \begin{pmatrix} \mathbf{W}[\![q, v, \underline{\overline{y}}, a', a]\!] \\ 0, 0 \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} \in \mathbb{R}^D \tag{82}$$

where $[\![q, v, \underline{\overline{y}}, a', a]\!] \in \{0,1\}^{|Q||\Gamma||\overline{\Sigma}_\varepsilon||A||A|}$ denotes the one-hot encoding of the tuple $(q, v, \underline{\overline{y}}, a', a)$,

$$\mathbf{W} \stackrel{\text{def}}{=} \left(\mathbf{W}^Q; \mathbf{W}^\Gamma; \mathbf{W}^A; \mathbf{A}'; \mathbf{A}\right) \in \mathbb{R}^{D \times \left(|Q||\Gamma||\overline{\Sigma}_\varepsilon||A||A|\right)} \tag{83}$$

for matrices $\mathbf{W}^Q, \mathbf{W}^\Gamma, \mathbf{W}^A$ *from Lemma F.4,* $\mathbf{A}', \mathbf{A} \in \mathbb{R}^{1\times D}$ *defined as*[32]

$$\mathbf{A}'_{1,(q,v,\underline{\overline{y}},a',a)} \stackrel{\text{def}}{=} a', \qquad q \in Q, v \in \Gamma, \underline{\overline{y}} \in \underline{\overline{\Sigma}}_\varepsilon, a, a' \in A \tag{84a}$$

$$\mathbf{A}_{1,(q,v,\underline{\overline{y}},a',a)} \stackrel{\text{def}}{=} a, \qquad q \in Q, v \in \Gamma, \underline{\overline{y}} \in \underline{\overline{\Sigma}}_\varepsilon, a, a' \in A, \tag{84b}$$

and $D \stackrel{\text{def}}{=} [|Q| + |\Gamma| + |A| + 2] + 2 + [1 + |\Gamma|] + 4$. *Then, it holds that*

$$\mathbf{r}_\Pi\left(y, t\right) = \begin{pmatrix} [\![q_t]\!] \\ [\![v]\!] \\ [\![a]\!] \\ a', a \\ 0, 0 \\ 0, \mathbf{0}_{|\Gamma|} \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix} \tag{85}$$

Proof. The first part (computing the one-hot encodings of $\sim_\delta$, v, and a') follows from the construction of the matrices $\mathbf{W}^Q, \mathbf{W}^\Gamma$, and $\mathbf{W}^A$ from Lemma F.4. The second part (computing the values a' and a with the matrices $\mathbf{A}'$ and $\mathbf{A}$) follows from the definition of the matrices $\mathbf{A}$ and $\mathbf{A}'$: By construction, the values of a' and a will be copied from the one-hot encodings into the resulting entry of the vector. ■

We now describe the two layers of the transformer that compute the quantities required to be able to determine the current configuration of the PTM at each time step. Here, we heavily rely on the construction by Pérez et al. (2021).

Let $\mathcal{M} = (Q, \Sigma, \Gamma, \delta_\mathcal{M}, q_\iota, q_\varphi)$ be a rationally-weighted PTM. Let $t \in \mathbb{N}$ and define

$$c\left(t\right) \stackrel{\text{def}}{=} \sum_{j=0}^{t-1} a_j \tag{86}$$

as the position of the head of the PTM at time t. Further, define the set

$$L\left(t\right) \stackrel{\text{def}}{=} \{j \mid c\left(j\right) = c\left(t\right), j \in [t-1]\}. \tag{87}$$

$L\left(t\right)$ contains the time steps at which the PTM $\mathcal{M}$ so far visited (and thus wrote to) the tape cell read by $\mathcal{M}$ at time t. Then, define $\ell\left(t\right)$ as

$$\ell\left(t\right) \stackrel{\text{def}}{=} \begin{cases} \max L\left(t\right) & \textbf{if } |L\left(t\right)| > 0 \\ t & \textbf{otherwise}. \end{cases} \tag{88}$$

In words, $\ell\left(t\right)$ denotes the time step at which the PTM $\mathcal{M}$ last visited (and thus wrote on) the tape cell read by $\mathcal{M}$ at time t if this cell was visited yet. Otherwise, $\ell\left(t\right)$ equals t.

Now, define the BOS symbol over the augmented alphabet Γ as

$$\text{BOS} \stackrel{\text{def}}{=} (q_0, \perp, \text{BOS}, 0, 0) \in Q \times \Gamma \times \underline{\overline{\Sigma}}_\varepsilon \times A \times A. \tag{89}$$

Let $q_\iota, q_1, \ldots, q_T$, $\perp, v_1, \ldots, v_T$, BOS, $y_1, \ldots, y_T$, and $a_0, a_1, \ldots, a_T$ be the sequences of states visited, symbols written on the processing tape, symbols written on the output tape, and actions performed by $\mathcal{M}$ in the first T steps. Define $a_{-1} = a_0 \stackrel{\text{def}}{=} 0$,

$$\mathbf{x}_t^0 \stackrel{\text{def}}{=} \begin{cases} \mathbf{r}_\Pi\left(\text{BOS}, t\right) & \textbf{if } t = 0 \\ \mathbf{r}_\Pi\left(\left(q_t, v_t, y_t, a_{t-1}, a_{t-2}\right), t\right) & \textbf{otherwise}, \end{cases} \tag{90}$$

[32] Recall that we can identify the actions of the PTM with the integers -1, 0, and 1.